

Practice Questions

LPP

Type-1: SIMPLEX METHOD

1. Maximize $z = x_1 + 4x_2$ Subject to $2x_1 + x_2 \leq 3, 3x_1 + 5x_2 \leq 9, x_1 + 3x_2 \leq 5, x_1, x_2 \geq 0$
2. Maximize $z = 100x_1 + 40x_2$
Subject to $10x_1 + 4x_2 \leq 2000, 3x_1 + 2x_2 \leq 900, 6x_1 + 12x_2 \leq 3000, x_1, x_2 \geq 0$
3. Maximize $z = x_1 + 3x_2$ Subject to $x_1 + 2x_2 \leq 10, 0 \leq x_1 \leq 5, 0 \leq x_2 \leq 4$
4. Maximize $z = 6x_1 - 2x_2 + 3x_3$ Subject to $2x_1 - x_2 + 2x_3 \leq 2, x_1 + 4x_3 \leq 4, x_1, x_2, x_3 \geq 0$
5. Maximize $z = 3x_1 + 2x_2$ Subject to $3x_1 + 2x_2 \leq 18, 0 \leq x_1 \leq 4, 0 \leq x_2 \leq 6, x_1, x_2 \geq 0$
6. Maximize $z = 3x_1 + 2x_2$ Subject to $x_1 + x_2 \leq 4, x_1 - x_2 \leq 2, x_1, x_2 \geq 0$
7. Minimize $z = x_1 - 3x_2 + x_3$
Subject to $3x_1 - x_2 + 3x_3 \leq 7, 2x_1 + 4x_2 \geq -12, -4x_1 + 3x_2 + 8x_3 \leq 10, x_1, x_2, x_3 \geq 0$
8. Maximize $z = 100x_1 + 50x_2 + 50x_3$
Subject to $4x_1 + 3x_2 + 2x_3 \leq 10, 3x_1 + 8x_2 + x_3 \leq 8, 4x_1 + 2x_2 + x_3 \leq 6, x_1, x_2, x_3 \geq 0$
9. Maximize $z = 4x_1 + 3x_2 + 6x_3$
Subject to $2x_1 + 3x_2 + 2x_3 \leq 440, 4x_1 + 3x_3 \leq 470, 2x_1 + 5x_2 \leq 430, x_1, x_2, x_3 \geq 0$
10. Maximize $z = 4x_1 + 10x_2$
Subject to $2x_1 + x_2 \leq 10, 2x_1 + 5x_2 \leq 20, 2x_1 + 3x_2 \leq 18, x_1, x_2 \geq 0$

TYPE-2: PENALTY METHOD

Using penalty (Big M) method to solve the following LPP

1. Maximize $z = 6x_1 + 4x_2$ Subject to $2x_1 + 3x_2 \leq 30, 3x_1 + 2x_2 \leq 24, x_1 + x_2 \geq 3, x_1, x_2 \geq 0$
2. Maximize $z = 4x_1 + 5x_2 + 2x_3$
Subject to $2x_1 + x_2 + x_3 \leq 10, x_1 + 3x_2 + x_3 \leq 12, x_1 + x_2 + x_3 = 6, x_1, x_2, x_3 \geq 0$
3. Minimize $z = x_1 + x_2$ Subject to $2x_1 + x_2 \geq 4, x_1 + 7x_2 \geq 7, x_1, x_2 \geq 0$
4. Minimize $z = 3x_1 + 8x_2$ Subject to $x_1 + x_2 = 200, x_1 \leq 80, x_2 \geq 60, x_1, x_2 \geq 0$
5. Maximize $z = 2x_1 + x_2 + 3x_3$ Subject to $x_1 + x_2 + 2x_3 \leq 5, 2x_1 + 3x_2 + 4x_3 = 12, x_1, x_2, x_3 \geq 0$
6. Maximize $z = 3x_1 - x_2$ Subject to $2x_1 + x_2 \geq 2, x_1 + 3x_2 \leq 3, x_2 \leq 4, x_1, x_2 \geq 0$
7. Maximize $z = 3x_1 - x_2$ Subject to $2x_1 + x_2 \leq 2, x_1 + 3x_2 \geq 3, x_2 \leq 4, x_1, x_2 \geq 0$
8. Maximize $z = 6x_1 + 4x_2$ Subject to $2x_1 + 3x_2 \leq 30, 3x_1 + 2x_2 \leq 24, x_1 + x_2 \geq 3, x_1, x_2 \geq 0$
Is the solution unique? If not, find another solution.
9. Minimize $z = 2x_1 + 3x_2$ Subject to $x_1 + x_2 \geq 5, x_1 + 2x_2 \geq 6, x_1, x_2 \geq 0$
10. Maximize $z = x_1 + 2x_2 + 3x_3 - x_4$
Subject to $x_1 + 2x_2 + 3x_3 = 15, 2x_1 + x_2 + 5x_3 = 20, x_1 + 2x_2 + x_3 + x_4 = 10, x_1, x_2, x_3, x_4 \geq 0$

TYPE-3 BASIC SOLUTIONS

Find all basic solutions to the following problem. Which of them are basic feasible, non-degenerate, infeasible basic and optimal feasible solutions?

1. Maximize $z = x_1 + 3x_2 + 3x_3$
Subject to $x_1 + 2x_2 + 3x_3 = 4, 2x_1 + 3x_2 + 5x_3 = 7, x_1, x_2, x_3 \geq 0$
2. Maximize $z = x_1 - 2x_2 + 4x_3$
Subject to $x_1 + 2x_2 + 3x_3 = 7, 3x_1 + 4x_2 + 6x_3 = 15, x_1, x_2, x_3 \geq 0$
3. Maximize $z = x_1 + x_2 + 3x_3$
Subject to $x_1 + 2x_2 + 3x_3 = 9, 3x_1 + 2x_2 + 2x_3 = 15, x_1, x_2, x_3 \geq 0$
4. Maximize $z = 2x_1 - 2x_2 + 4x_3 - 5x_4$
Subject to $x_1 + 4x_2 - 2x_3 + 8x_4 = 2, -x_1 + 2x_2 + 3x_3 + 4x_4 = 1, x_1, x_2, x_3, x_4 \geq 0$
5. Maximize $z = 2x_1 + 3x_2 + 4x_3 + 7x_4$
Subject to $2x_1 + 3x_2 + x_3 + 4x_4 = 8, x_1 - 2x_2 + 6x_3 - 7x_4 = -3, x_1, x_2, x_3, x_4 \geq 0$
6. Maximize $z = 2x_1 + 3x_2$
Subject to $x_1 + 3x_2 \leq 6, 3x_1 + 2x_2 \leq 6, x_1, x_2 \geq 0$

Covert the following LPP to standard form

7. Maximize $z = 3x_1 + 5x_2$
subject to $3x_1 + 2x_2 \leq 15, 2x_1 + 5x_2 \geq 12, x_1, x_2 \geq 0$
8. Minimize $z = -3x_1 + 2x_2 - x_3$
Subject to $x_1 - 3x_2 + 2x_3 \geq -6, 3x_1 + 4x_3 \leq 3, -3x_1 + 5x_2 \leq 4, x_1, x_2 \geq 0, x_3$ is unrestricted
9. Maximise $z = 2x_1 - x_2 + x_3$
Subject to $x_1 + x_2 \leq 10, 2x_1 - x_2 - x_3 \geq -4, x_2 - x_1 \leq 7, x_1, x_2 \geq 0$
10. Maximize $z = 3x_1 + 4x_2 - 2x_3$
Subject to $6x_1 - 4x_2 \leq 5, 3x_1 + x_2 + 4x_3 \geq 11, 4x_1 + 3x_2 \leq 2, x_1 \geq 0, x_2 \geq 0$
11. Maximize $z = 3x_1 + 2x_2 + 5x_3$
Subject to $2x_1 - 3x_2 \leq 3, x_1 + 2x_2 + 3x_3 \geq 5, 3x_1 + 2x_3 \leq 2, x_1, x_2 \geq 0$
12. Minimize $z = 2x_1 + 3x_2$
Subject to $2x_1 - 3x_2 - x_3 = -4, 3x_1 + 4x_2 - x_4 = -6, 2x_1 + 5x_2 + x_5 = 10, 4x_1 - 3x_2 + x_6 = 18$
 $x_3, x_4, x_5, x_6 \geq 0$
13. Minimize $z = x_1 + 2x_2 + 3x_3$
Subject to $3x_1 + 4x_2 \leq 5, 5x_1 + x_2 + 6x_3 = 7, 8x_1 + 9x_2 \leq 9, x_1, x_2, x_3 \geq 0$
Also put the problem in matrix form.
14. Maximize $z = 2x_1 + 3x_2 + 6x_3$
Subject to $3x_1 - 2x_2 + 4x_3 \leq 5, 2x_1 + 5x_2 = 10, x_1 + 2x_2 + x_3 \leq 2, x_1, x_2, x_3 \geq 0$
Also put the problem in matrix form.

Type-4 : Duality

Write the duals of the following LPP

1. Maximize $z = 3x_1 + 5x_2$
Subject to $2x_1 + 6x_2 \leq 50, 3x_1 + 2x_2 \leq 35, 5x_1 - 3x_2 \leq 10, x_2 \leq 20, x_1 \geq 0$
2. Maximize $z = 2x_1 + x_2 + x_3$
Subject to $x_1 + x_2 + x_3 \geq 6, 3x_1 - 2x_2 + 3x_3 = 3, -4x_1 + x_3 \leq 10, x_1, x_3 \geq 0, x_2$ unrestricted
3. Minimize $z = x_2 + 3x_3$
Subject to $2x_1 + x_2 \leq 3, x_1 + 2x_2 + 6x_3 \geq 5, -x_1 + x_2 + 2x_3 = 2, x_1, x_2, x_3 \geq 0$
4. Minimize $z = 3x_1 - 2x_2 + x_3$
Subject to $2x_1 - 3x_2 + x_3 \leq 5, 4x_1 - 2x_2 \geq 9, -8x_1 + 4x_2 + 3x_3 = 8, x_1, x_2 \geq 0, x_3$ unrestricted
5. Maximize $z = 3x_1 + 17x_2 + 9x_3$
Subject to $x_1 - x_2 + x_3 \geq 3, -3x_1 + 2x_3 \leq 1, 2x_1 + x_2 - 5x_3 = 1, x_1, x_2, x_3 \geq 0$
6. Maximize $z = 2x_1 - x_2 + 3x_3$
Subject to $x_1 - 2x_2 + x_3 \geq 4, 2x_1 + x_3 \leq 10, x_1 + x_2 + 3x_3 = 20, x_1, x_3 \geq 0, x_2$ unrestricted
7. Minimize $z = x_1 - 3x_2 - 2x_3$
Subject to $3x_1 - x_2 + 2x_3 \leq 7, 2x_1 - 4x_2 \geq 12, -4x_1 + 3x_2 + 8x_3 = 10, x_1, x_2 \geq 0, x_3$ unrestricted
8. Maximize $z = x_1 - 2x_2 + 3x_3$
Subject to $-2x_1 + x_2 + 3x_3 = 2, 2x_1 + 3x_2 + 4x_3 = 1, x_1, x_2, x_3 \geq 0$